EE3-10
EE9-CS3-1

IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2015

MSc and EEE/EIE PART III/IV: MEng, BEng and ACGI

Corrected copy

MATHEMATICS FOR SIGNALS AND SYSTEMS

Monday, 7 December 9:00 am

Time allowed: 3:00 hours

There are THREE questions on this paper.

Answer ALL questions. All questions carry equal marks.

NO Calculators allowed

Any special instructions for invigilators and information for candidates are on page 1.

Examiners responsible First Marker(s) : P.T. Stathaki
Second Marker(s) : D. Angeli

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1. a) Specify whether the matrix below has an inverse without attempting to compute the inverse.

$$R=\begin{bmatrix} -1 & 1 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 & 0 \\ 0 & 0 & 1 & -1 & 0 \\ 0 & 0 & 1 & 0 & 1 \\ 1 & -1 & 1 & 1 & 0 \end{bmatrix}$$

[2]

b) Let $A=\begin{bmatrix} 1 & a & 0 \\ 1 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$. Determine those values of a for which A is invertible. **[2]**

c) Find the volume of the parallelepiped S formed by the triple of vectors in R^3, $x=(1,1,1)^T$, $y=(2,3,4)^T$, $z=(1,1,5)^T$. **[2]**

d) An $n\times n$ matrix A is called skew-symmetric if $A^T=-A$. Show that if A is skew-symmetric and n is an odd positive integer, then A is not invertible. [Hint: I would suggest you to work with determinants] **[4]**

e) Let $A=\begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}$ and assume that $\det(A)=10$. Find $\det(5A)$, $\det(3A^{-1})$, $\det(3A^3)$, $\det[2(A^T)^{-1}]$, and $\det(B)$ with $B=\begin{bmatrix} a & g & d \\ b & h & e \\ c & i & f \end{bmatrix}$. **[4]**

f) Find the solution of the following system of equations using QR decomposition.

$$\begin{aligned} x-2y-2z&=3 \\ -x+2y+3z&=1 \\ 2x-2y-2z&=-2 \end{aligned}$$

The third column of Q is $\begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \\ 0 \end{bmatrix}$. **[6]**

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2. a) Consider the matrix

$$A = \begin{bmatrix} 1 & 1 & -1 & 2 \\ 2 & 1 & -2 & 2 \\ 1 & 0 & -1 & 0 \end{bmatrix}$$

(i) By using elimination find the dimension and a basis of the row space of A, $R(A)$. **[2]**

(ii) Find the dimension and a basis of the nullspace of A, $N(A)$. **[2]**

(iii) Find the dimension and a basis of the column space of A, $C(A)$. **[2]**

(iv) Find the dimension and a basis of the left nullspace of A, $N(A^T)$. **[2]**

b) Let S be a set of m vectors in R^n. Mark each statement (i)-(v) **True** or **False**. Justify your answer.

(i) If $m > n$ then the vectors in S are linearly independent. **[1]**

(ii) If the zero vector is in S, then the vectors in S are linearly dependent. **[1]**

(iii) If the vectors in S are linearly independent and T is a subset of S, then the vectors in T are linearly independent. **[1]**

(iv) If the vectors in T are linearly dependent and T is a subset of S, then the vectors in S are linearly dependent. **[1]**

(v) The linear system $Ax = b$, with A being a matrix and x, b being column vectors, has a unique solution if and only if the column vectors of A are linearly independent. The dimension of A is not specified. **[2]**

c) We are seeking to fit the 4 two-dimensional points $(-2,0), (-1,0), (0,1), (1,1)$ onto a straight line.

(i) Give the system of equations that we must solve in order to achieve the above requirement. Explain why the system doesn't have a solution. **[2]**

(ii) Find an approximate solution of the system using the least squares approach and give the equation for the required straight line. **[2]**

(iii) Calculate the magnitude of the error of the approximation. **[2]**

3. a) Consider a matrix A with characteristic polynomial $\lambda^5 - 10\lambda^4 + 3\lambda^3 + \lambda^2 - 2\lambda + 7$. Is A invertible? Justify your answer. **[1]**

b) Suppose that $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$ is an eigenvector of a matrix A corresponding to an eigenvalue of 3 and $\begin{bmatrix} 2 \\ 1 \end{bmatrix}$ is an eigenvector of A corresponding to an eigenvalue of -2. Compute $A^3\begin{bmatrix} 4 \\ 3 \end{bmatrix}$.

[4]

c) Consider the matrix A:

$$A = \begin{bmatrix} 4 & 0 & 0 & 0 \\ 0 & 4 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 1 & 0 & 0 & 2 \end{bmatrix}$$

Determine if A is diagonalizable, and if so, diagonalize it. **[5]**

d) Consider the matrices A and B shown below.

$$A = \begin{bmatrix} 1 & 3 & 3 \\ -3 & -5 & -3 \\ 3 & 3 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} 2 & 4 & 3 \\ -4 & -6 & -3 \\ 3 & 3 & 1 \end{bmatrix}$$

Both matrices have characteristic polynomial $p_A(\lambda) = p_B(\lambda) = -(\lambda - 1)(\lambda + 2)^2$.

(i) Find all eigenvectors of matrix A. **[3]**
(ii) Find all eigenvectors of matrix B. **[3]**
(iii) State which of the above matrices A, B are diagonalizable. **[2]**
(iv) Diagonalize the matrix or matrices, if any, stated in (iii) above. **[2]**

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